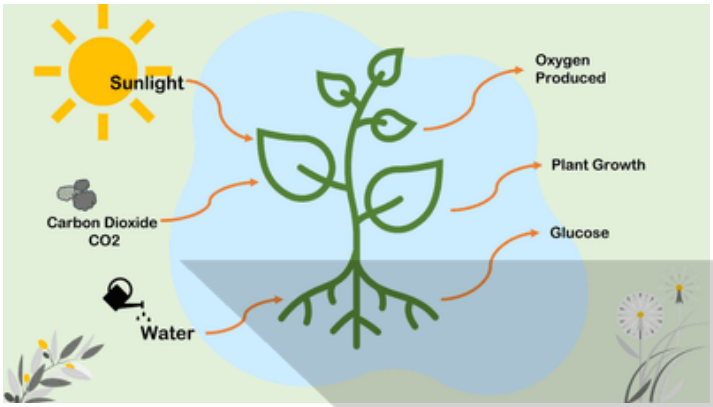
1. Plant Nutrition

1.1. Photosynthesis

Photosynthesis: the process by which plants manufacture carbohydrates from raw materials using energy from light.



$$Carbon Dioxide + Water \xrightarrow{light+chlorophyll} Glucose + Oxygen$$

$$6CO_2 + 6H_2O \xrightarrow{light+chlorophyll} C_6H_{12}O_6 + 6O_2$$

- Only **EXTENDED Students** must know the **Balanced Chemical Equation**
- The carbon dioxide diffuses through the open stomata of a plant leaf, and water is taken up through the roots.
- Chlorophyll is a **green pigment** that traps light energy and converts it into chemical energy to form carbohydrates and their subsequent storage.
- Glucose is used for respiration, energy storage, cellulose cell walls, and making proteins and sugars.

Use and Storage of the Carbohydrates Made in Photosynthesis

- starch as an energy store
- cellulose to build cell walls
- glucose used in respiration to provide energy
- sucrose for transport in the phloem

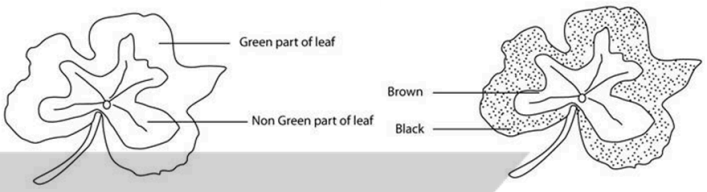
1.2. Mineral Requirements

Nitrate ions	Magnesium ions
Making amino acids	Making chlorophyll
Deficiency: small plant due to slow/stunted growth	Deficiency: The plant lacks chlorophyll, leaves turn yellow.

Tip! You need to know the purpose of these required nutrients.

1.3. Investigation of Chlorophyll

- Take a potted plant with variegated (green and white) leaves.
- De-starch the plant by keeping it in complete darkness for about 48 hours.
- Expose the plant to sunlight for a few days.
- Leaf boiled in water for 2 minutes to break down cell walls, denature enzymes and allow for easier penetration by ethanol.
- Warmed in ethanol until the leaf is colourless to extract chlorophyll, which would mask the observation
- Dipped into the water briefly: to help soften the leaf
- The leaf is placed on a white tile, and iodine is added. If starch is present, the colour will be blue-black; if absent, it will remain brown.

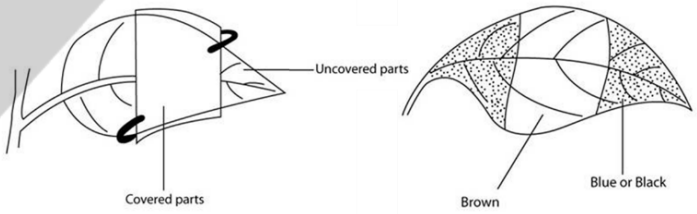


(a) variegated leaf before (b) after testing for starch

1.4. Investigation of Light Intensity

NOTE: This type of question is frequently asked in Paper 2.

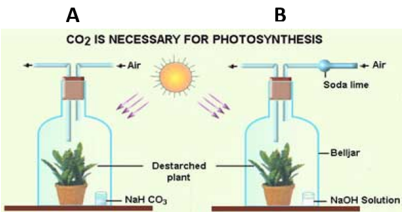
- De-starch the plant by keeping it in the dark for 48 hours
- Place a stencil over part of a leaf
- Place the leaf in sunlight for 4-6 hours
- Remove the stencil and test for starch
- **+ve result = parts which received light turn blue-black**
- **-ve result = parts that didn't receive light remain brown**



(a) leaf covered with stencil (b) after testing for starch

1.5. Investigation of Carbon Dioxide Concentration

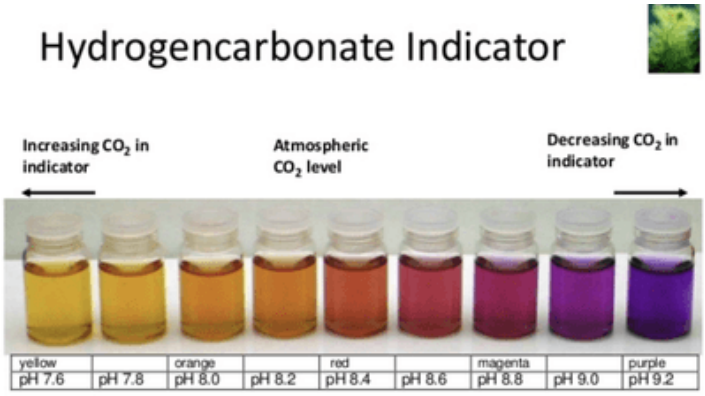
- Take two de-starched potted plants.
- Cover both the plants with bell jars and label them A and B.
- Inside A, keep $NaHCO_3$ (Sodium Bicarbonate). It produces CO_2 .
- Inside B, keep $NaOH$ (Sodium Hydroxide). It absorbs CO_2 .
- Keep both set-ups and do the starch test in the sunlight for at least 6 hours



- The leaves of Plant A will turn black after the starch test
- The leaves of Plant B will remain brown after the starch test

Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by at Enderun Colleges on 05/04/26.

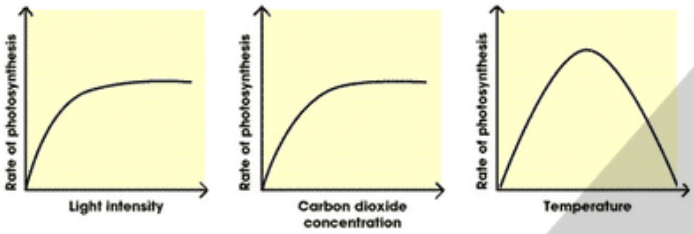
Hydrogencarbonate indicator: measures the carbon dioxide concentration



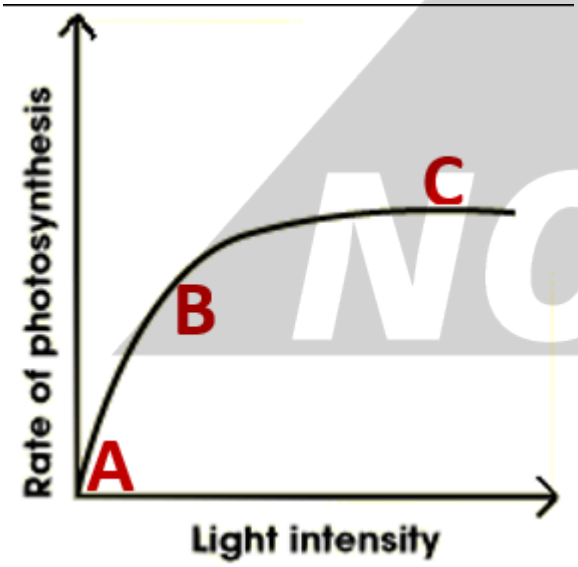
Key: PMROY, purple being the lowest carbon dioxide concentration, and yellow the highest.

1.6. Limiting Factors

Limiting Factors: Conditions where something present in the environment (Temperature, Carbon Dioxide Concentration and Light Intensity) is in such short supply that it restricts life processes.



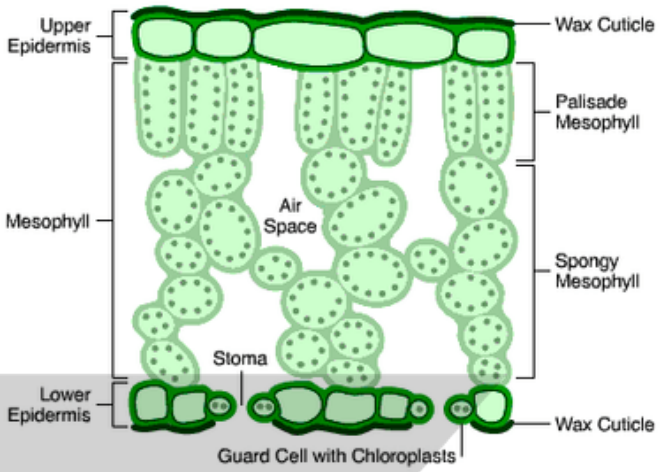
Light intensity



- As the amount of light increases, the rate of photosynthesis increases (A-B)
- The limiting factor is light
- Increasing the amount of light after a certain point does not affect the rate (C)
- The limiting factor is now carbon dioxide or temperature.

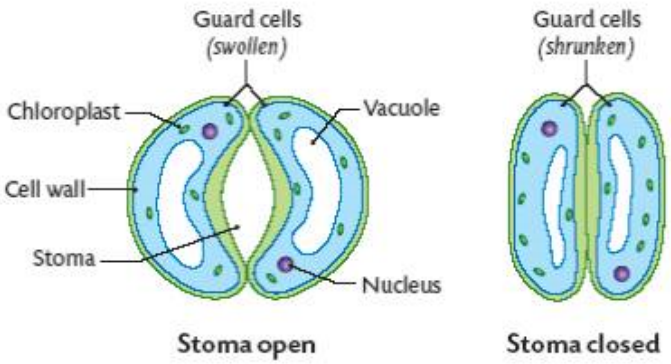
1.7. Leaf Structure

Most dicotyledonous plant leaves have a **large surface area** and are **thin**.



Structure	Function
Waxy Cuticle	A waterproof waxy layer that prevents water loss from the top of the leaf
Upper & Lower Epidermis	Thin and transparent to allow light to enter the palisade mesophyll cells
Palisade Mesophyll Cells	Found at the top of the cell and contains the **MOST chloroplasts** that absorb sunlight for photosynthesis.
Spongy Mesophyll Cells	Irregularly shaped cells create air spaces to allow the gaseous exchange to take place; do not contain many chloroplasts
Vascular Bundle	Made up of xylem and phloem
Xylem	A tissue vessel that transports water and mineral ions
Phloem	A tissue vessel that transports sucrose and amino acids
Stomata/Stoma	Tiny gaps that open and close to allow the gaseous exchange to occur
Guard Cells	Control the opening and closing of stomata

Opening and closing of Stomata:



Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by at Enderun Colleges on 05/04/26.

- Stomata open when guard cells absorb water by osmosis, become turgid, and curve outward, creating a gap.
- They close when guard cells lose water, become flaccid, and collapse, sealing the pore to reduce water loss.

Syllabus 6.2.3: You must be able to explain how the structures above adapt leaves for photosynthesis

1.8. Adaptations of Leaf Structure for Photosynthesis

- Here is a table of different adaptations of the specific leaf structure helping towards photosynthesis.

Adaptation	Functions
Large Surface Area of Leaf	Increase surface area for diffusion of carbon dioxide and absorption of light for photosynthesis.
Thin	Allow carbon dioxide to diffuse quickly into the palisade mesophyll cells
Chlorophyll	Absorb light energy so photosynthesis can take place.
Network of Veins	Allow efficient transport of water and mineral ions throughout the plant
Epidermis is thin	Allow more light to reach the palisade mesophyll cells
Stomata/Stoma	Allow oxygen and carbon dioxide to continuously diffuse in and out of the leaf

Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by at Enderun Colleges on 05/04/26.